

Decent Coaching Classes / VIII NCERT / Worksheet / Factorization

Factorise the following expressions completely. Show all steps.

Section A: Factorisation by Regrouping

Q1. $2xy + 3 + 2y + 3x$

Q2. $6xy - 4y + 6 - 9x$

Q3. $ax + bx - ay - by$

Q4. $3ax - 6ay - 8by + 4bx$

Section B: Factorisation of Quadratic Expressions

Q5. $x^2 + 7x + 12$

Q6. $y^2 - 9y + 20$

Q7. $m^2 + 2m - 15$

Q8. $p^2 - 5p - 24$

Answer Key & Solutions

Q1. $2xy + 3 + 2y + 3x$

$$\begin{aligned} 2xy + 3 + 2y + 3x &= 2xy + 2y + 3x + 3 \\ &= 2y(x + 1) + 3(x + 1) \\ &= (x + 1)(2y + 3) \end{aligned}$$

Q2. $6xy - 4y + 6 - 9x$

$$\begin{aligned} 6xy - 4y + 6 - 9x &= 6xy - 4y - 9x + 6 \\ &= 2y(3x - 2) - 3(3x - 2) \\ &= (3x - 2)(2y - 3) \end{aligned}$$

Q3. $ax + bx - ay - by$

$$\begin{aligned} ax + bx - ay - by &= x(a + b) - y(a + b) \\ &= (a + b)(x - y) \end{aligned}$$

Q4. $3ax - 6ay - 8by + 4bx$

$$\begin{aligned} 3ax - 6ay - 8by + 4bx &= 3ax - 6ay + 4bx - 8by \\ &= 3a(x - 2y) + 4b(x - 2y) \\ &= (x - 2y)(3a + 4b) \end{aligned}$$

Q5. $x^2 + 7x + 12$

Need two numbers with product 12 and sum 7: 3 and 4

$$\begin{aligned} x^2 + 7x + 12 &= x^2 + 3x + 4x + 12 \\ &= x(x + 3) + 4(x + 3) \\ &= (x + 3)(x + 4) \end{aligned}$$

Q6. $y^2 - 9y + 20$

Need two numbers with product 20 and sum -9: -4 and -5

$$\begin{aligned} y^2 - 9y + 20 &= y^2 - 4y - 5y + 20 \\ &= y(y - 4) - 5(y - 4) \\ &= (y - 4)(y - 5) \end{aligned}$$

Q7. $m^2 + 2m - 15$

Need two numbers with product -15 and sum 2: 5 and -3

$$\begin{aligned} m^2 + 2m - 15 &= m^2 + 5m - 3m - 15 \\ &= m(m + 5) - 3(m + 5) \\ &= (m + 5)(m - 3) \end{aligned}$$

Q8. $p^2 - 5p - 24$

Need two numbers with product -24 and sum -5: -8 and 3

$$\begin{aligned} p^2 - 5p - 24 &= p^2 - 8p + 3p - 24 \\ &= p(p - 8) + 3(p - 8) \\ &= (p - 8)(p + 3) \end{aligned}$$